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VMM Device Test for Continous Flexray & CAN ECU

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Issued by **zhangzhaomin**	Approved by **zhangzhaomin**

DESIGN VERIFICATION METHOD
Vehicle Mode Management

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0 CHANGE LOG

0.1 List of Test Cases

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0.2 Test Cases Change Log

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CarMode Factory 17
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CarMode Transport 12
ECU recovery mechanism when High and Low Power supply 40
ECU shoud have timeout mechanism when NM frame is sent with PNC is 0 more than 15s 48
ECU shoud have timeout mechanism when ECU just send APP frame and do not send NM frame 50
ECU shoud have timeout mechanism and ECU should reset when RG (Resource group), PNC can not set 0 45
ECU shoud have timeout mechanism and ECU should reset when Vote and PNC can not set 0 on time 42
Energy Consumption During vehicle Start 27
Master ECU reset when Quiescent current monitor is active 53
VMM ECU DID 55

0.2.2 Updated Test Cases in this issue

HW input wake up sources Pin XX 31
Internal timer wake up sources Timer XX 34
Sensor signal input wake up sources Signal XX 36
Test for not wake up sources Pin XX 30
Wireless wakeup sources Wireless XX 38

0.2.3 Deleted Test Cases in this issue

Network wakeup sources Frame XX

1 Purpose

This test code will include some testing/ verification of Vehicle Mode Management (VMM) i.e. system solution of the power design in the car. The device DVM tests shall be done in test bench or equivalent.

The purpose is to verify given requirements and shall be a part of the development process. The feedback of what the person experience during the tests is as important as the test itself.

This DVM is aimed at All ECUs, and the device level DVMs are performed by the supplier.

2 Equipment

2.1 Equipment for Component Test

Note that all specified equipment may not be needed for all tests and some other equipment may be needed for others.

The equipment used and set-up shall be documented, e.g. schematics/photos/instrument names/manufacturer/numbers/calibration etc. The documentation of set-up and instrument shall be part of the report.

Equipment	Usage
Test PC	Runs SW applications
CANalyzer card including cables (LapCan card) or equivalent equipment to facilitate tests	HW used to view application data on a CAN Network
DSA (Diagnostic and SWDL Application)	Application used to perform diagnostic services
Voltage measurement tool, calibrated for test range	measure voltage supply
Current measurement tool, calibrated for test range	Measure the current consumption of the component

3 Preparations/Test Arrangement

Test Preparations

Need to list the ECU all wake up trigger sources, and list each trigger sources valid and invalid value.
Then select corresponding general test cases to do test for each trigger source.

General test info

3.1 Current consumption measurement

Some guidelines on how to do the current consumption measurements are described below:

- The current consumption shall be an average value of the measured current consumption during the time measured. The maximum current level shall also be measured.
- All measurements shall be done at voltage level 12.5 V ± 0.25 V.
- The measured value shall have an accuracy of ± 5% or ± 25 µA.
- Measurement at 26±5°C.
- Make sure all relevant connections on the components are in place. This includes sensors and external loads. External loads, such as fans/heaters/pumps etc controlled by the ECU, functionality shall follow the ECU functionality, e.g. then the ECU is sleeping the external loads should not draw current.
- All measurements shall be started when the component is powerfeed. When the current is stable the actual measurement shall take place; the average and maximum value shall be evaluated during one minute. The time for stable current consumption varies for different components and modes and may in some cases take up to 30 minutes.
- Measurements shall be done in such a way that all contribution to total current consumption is captured i.e. current through potential inputs/outputs shall be captured. this is normally done on + or - feeding of the component but may differ depending on design. Can be necessary to measure both + and - feeding to ensure no leakage currents.

3.2 Signal mapping

Vehicle Mode signals mentioned in the test procedures shall be mapped to the signals described below:

- UsageMode: UsgModSts
- CarMode (main and subtype): CarModSts1, CarModSubtypWdCarModSubtyp
- Energy Consumption Fault : FltEgyCnsWdSts
- ElPowerLevel (main and subtype): PwrLvlElecMai, PwrLvlElecSubtyp
- ElEnergyLevel (main and subtype): EgyLvlElecMai, EgyLvlElecSubtyp

These signals shall be sent in a system signal group record: VehModMngtGlbSafe1.

If ECU comply with AutoSAR 4.03 or later PNC activation shall be used.

3.2.1 Default signal setting

If nothing else is stated:

UsageMode; Active: `UsgModSts=Active {0b1011}`

CarMode; Normal;Normal: `CarModSts1=Normal {0b000}, CarModSubtypWdCarModSubtyp=Normal {0b000}`

ElPowerLevel; Default: `PwrLvlElecMai= 0 {0x0000}, PwrLvlElecSubtyp= 0 {0x0000}`

Fault detection of high current consumption, Default: FltEgyCns = No_Flt

3.3 Test Procedure

Explanation of the test procedures:

- Objective: A short description of what is tested.
- Test strategy: A short description of how the test is performed.
- Preconditions and information: Which actions to perform prior to the test.
- Procedure: Step by step instructions of how the test is performed.
- Test step: One step of the procedure.
- Expected response: After each action you can see what is supposed to happen.
- Observed response: After each action you can comment the actual response.

3.4 Simulation

For all device tests (stand alone) all test environment signals and their update bits must be simulated or sent by test equipment to the connected ECU (e.g. CarMode signal and UsageMode signal etc).

4 Test Methods

This chapter contains the test scope of this DVM. The test scope contains all test specifications, test cases and links to requirements.

All tests shall be done in:

1- Test bench or set up with relevant environmental surroundings, e.g. signal interface and electrical feeding.
N.B: It may be necessary to resend the PNC mapped to the VMM signal group to activate ECU before sending out other signals.

The VMM signals, See 3.2 shall be mapped to internal ECU states. The specific internal states are not described in this document - the specific internal states should be part of the ECU implementation requirements.

References in SystemWeaver model

Test scope: VMM Device test Continous Flexray & CAN

Version:(4) Status:CS Released

Test specification summary	Ver	Status
Current Consumption and ECU Behaviour continous Flexray & CAN	(5)	CS Released

4.1 Test Specification: Current Consumption and ECU Behaviour continous Flexray & CAN (Ver: 5)

Test Specification Description :
The purpose of this DVM is to verify ECU behaviour and Current Consumption in different ECU states.

This DVM is aimed at ECUs connected to Flexray and/or CAN networks with a continous power supply.

4.1.1 Test case : CarMode Normal (Ver: 6)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10678
Legacy ID:
Test Environment:

Test case Description:

Purpose:
Objective:

Test the functionality and quiescent current consumption in CarMode = NORMAL.

Test Strategy:

- **All measurements shall be started when the component is powerfeed.**

  When the current is stable the actual measurement shall take place; the average and maximum value shall be evaluated during one minute.

  The time for stable current consumption varies for different components and modes and may in some cases take up to 30 minutes.

- All loads connected to the ECU, e.g. external sensors, shall be included during measurements.

Requirements:

Test Script Description :

Step	Action	Expected Result
1	PowerFeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000}	CarMode Normal; Normal is sent out on the Networks.
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
6	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
7	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
8	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated after UsageMode Abandoned is sent out.
9	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired.
10	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
11	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
12	Measure the current consumption. (Measure in CarMode Normal	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU.

Step	Action	Expected Result
	;Normal & UsageMode Abandoned) Fill in the measured value in the attached excel template under the current consumption sheet.	The measured current consumption shall be included in in the attached excel template.
13	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
14	Set UsageMode to Inactive UsgModSts = Inactive {0b0001}	UsageMode Inactive is sent out. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Inactive.
15	Deactivate the PNC and simulation of test environment signals on the bus..	The simulated network communication shall be deactivated after UsageMode Inactive is distributed.
16	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired.
17	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
18	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
19	Measure the current consumption. (Measure in CarMode Normal;Normal & UsageMode Inactive. Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
20	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
21	Set UsageMode to Convinience UsgModSts = Convinience {0b0010}	UsageMode convenience is sent out. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Convenience.
22	After sending out UsageMode Convenience, deactivate the PNC and simulation of test environment signals on	The simulated network communication shall be deactivated after UsageMode Convenience is distributed.

Step	Action	Expected Result
	the bus.	
23	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired.
24	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
25	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
26	Measure the current consumption. (Measure in CarMode Normal;Normal & UsageMode Convenience) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
27	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
28	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Active.
29	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
30	Measure the current consumption. (Measure in CarMode Normal;Normal & UsageMode Active) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
31	Set UsageMode to Driving UsgModSts = Driving {0b1101}	The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Driving.
32	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
33	Measure the current consumption.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU.

Step	Action	Expected Result
	(Measure in CarMode Normal;Normal & UsageMode Driving) Fill in the measured value in the attached excel template under the current consumption sheet.	The measured current consumption shall be included in in the attached excel template.
34	Check DTC to see if any is set due to this test.	No DTC's are set
35	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
36	Check for errorframes and fill in the amount of error frames detected in the attached excel template under the errorframe sheet. Fill in 0 if no were detected.	The number of error frames observed shall be in the attached excel template
37	Remove Powerfeed for the Component/s.	

4.1.2 Test case : CarMode Transport (Ver: 6)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10679
Legacy ID:
Test Environment:
Test Script Action:
Test Script
Expected Result:

Test case Description:

Purpose:
Objective:

Test the functionality and quiescent current consumption in CarMode = TRANSPORT.

Test strategy:

Preconditions and information:

- **All measurements shall be started when the component is powerfeed.**

  When the current is stable the actual measurment shall take place; the average and maximum value shall be evaluated during one minute.

  The time for stable current consumption varies for different components and modes and may in some cases take up to 30 minutes.

- All loads connected to the ECU, e.g. external sensors, shall be included during measurements.

Requirements:

Test Script Description :

Step	Action	Expected Result
1	Powerfeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal;Normal CarModSts1=Normal {0b000}, CarModSubtypWdCarModSubtyp=Normal0b000	CarMode Normal; Normal is sent out on Network.
5	Set UsageMode to Active UsgModSts=	UsageMode Active is sent out on Network.

Step	Action	Expected Result
	Active {0b1011}	The Tested ECU shall be set in required internal mode due to CarMode Normal,UsageMode Active
6	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
7	Set UsageMode Abandoned UsgModSts = Abandoned {0b000}	UsageMode Abandoned is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
8	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdC arModSubtyp = Transport {0b000}	CarMode Transport is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Transport UsageMode Abandoned
9	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
10	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired
11	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
12	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
13	Measure the current consumption. (Measure in CarMode Transport; Transport & UsageMode Abandoned) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
14	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
15	Set UsageMode to Driving UsgModSts = Driving {0b1101}	UsageMode Driving is sent out on the Network
16	Set CarMode Normal;Transport Driving CarModSts1 = Normal {0b000} CarModSubtypWdCarModSubtyp =	CarMode Normal;Transport Driving is sent out on the network. The tested ECU shall be set in required internal mode due

Step	Action	Expected Result
	Transport Driving {0b011}	to CarMode Normal;Transport Driving UsageMode Driving
17	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
18	Measure the current consumption. (Measure in CarMode Normal;Transport Driving & UsageMode Driving) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
19	Set UsageMode Abandoned UsgModSts = Abandoned {0b000}	UsageMode Abandoned is sent out on the Network The simulated network communication shall be deactivated after UsageMode Abandoned is sent out.
20	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdCarModSubtyp = Transport {0b000}	CarMode Transport is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Transport UsageMode Abandoned
21	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
22	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired.
23	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
24	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
25	Measure the current consumption. (Measure in CarMode Transport;Transport & UsageMode Abandoned) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
26	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The simulated network communication shall be activated.

Step	Action	Expected Result
27	Set UsageMode Inactive UsgModSts = Inactive {0b001}	UsageMode Inactive is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Transport UsageMode Inactive
28	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
29	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired.
30	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
31	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
32	Measure the current consumption. (Measure in CarMode Transport;Transport & UsageMode Inactive) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
33	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
34	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Transport UsageMode Active
35	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
36	Measure the current consumption. (Measure in CarMode Transport;Transport & UsageMode Active) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
37	Check DTC to see if any is set due to this test	No DTC's are set
38	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.

Step	Action	Expected Result
39	Check for errorframes and fill in the amount of error frames detected in the attached excel template under the errorframe sheet. Fill in 0 if no were detected.	The number of error frames observed shall be in the attached excel template
40	Remove powerfeed for the component/s	

4.1.3 Test case : CarMode Factory (Ver: 6)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10680
Legacy ID:
Test Environment:

Test case Description:

Purpose:
Objective:

Test the functionality and quiescent current consumption in CarMode = FACTORY.

Test strategy:

preconditions and information:

- **All measurements shall be started after the component is set to sleep and the current consumption is stable. This time is varying between different components and can in some cases take up to 30 minutes before the measurement can start**

  The current consumption shall then be measured during 1minute; average and maximum value shall be noted and compared to requirement..

- All loads connected to the ECU, e.g. external sensors, shall be included during measurements

Requirements:

Test Script Description :

Step	Action	Expected Result
1	Powerfeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal;Normal CarModSts1 = Normal{0b000}, CarModSubtypWdCarMo dSubtyp = Normal{0b000}	CarMode Normal;Normal is sent out on Network
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on Network The tested ECU shall be set in required internal mode due to CarMode Normal;Normal UsageMode Active

Step	Action	Expected Result
6	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
7	Set UsageMode to Abandoned UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network
8	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubtyp = Factory {0b000}	CarMode Factory is sent out on the network The tested ECU shall be set in required internal mode due to CarMode Factory;Factory UsageMode Abandoned
9	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated after CarMode Factory;Factory is sent out.
10	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired
11	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
12	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
13	Measure the current consumption. (Measure in CarMode Factory;Factory & UsageMode Abandoned) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
14	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
15	Set UsageMode Inactive UsgModSts = Inactive {0b001}	UsageMode Inactive is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Factory UsageMode Inactive
16	Set CarMode Normal;Factory Paused CarModSts1 = Normal {0b000}, CarModSubtypWdCarMo	CarMode Normal;Factory Paused is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Normal;Factory Paused UsageMode Inactive

Step	Action	Expected Result
	dSubtyp = Factory Paused {0b001}	
17	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
18	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired
19	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
20	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template
21	Measure the current consumption. (Measure in CarMode Normal;Factory Paused & UsageMode Inactive) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
22	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
23	Set UsageMode to Driving UsgModsts = Driving {0b1101}	UsageMode Driving is sent out on the Network
24	Set CarMode Normal;Factory Driving CarModSts1 = Normal {0b000}, CarModSubtyp = Factory Driving {0b010}	CarMode Normal; Factory Driving is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Normal;Factory Driving Usage Mode Driving
25	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
26	Measure the current consumption. (Measure in CarMode Normal;Factory Driving & UsageMode Driving Fill in the measured value in the attached excel template under the current	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.

Step	Action	Expected Result
	consumption sheet.	
27	Set UsageMode to Abandoned UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network
28	Set CarMode Factory; Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubtyp = Factory {0b000}	CarMode Factory;Factory is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Factory;Factory UsageMode Abandoned
29	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated
30	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired
31	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
32	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
33	Measure the current consumption. (Measure in CarMode Factory;Factory & UsageMode Abandoned) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template
34	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
35	Set UsageMode Inactive UsgModSts = Inactive {0b001}	The tested ECU shall be set in required internal mode due to CarMode Transport UsageMode Inactive.
36	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
37	Investigate if the component stop communicating after communication shut down timer (10 sec)?	Component stop communicating after communication shut down timer (10 sec) has expired
38	Does component go to sleep after component shutdown timer (15 sec)?	Component go to sleep after component shutdown timer (15 sec) has expired.
39	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template

Step	Action	Expected Result
40	Measure the current consumption. (Measure in CarMode Factory;Factory & UsageMode Inactive) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template
41	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
42	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on the Network
43	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template
44	Measure the current consumption. (Measure in CarMode Factory;Factory & UsageMode Active) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template
45	Check DTC to see if any is set due to this test	No DTC's are set
46	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
47	Check for errorframes and fill in the amount of error frames detected in the attached excel template under the errorframe sheet. Fill in 0 if no were detected.	The number of errorframes observed shall be in the attached excel template
48	Remove Powerfeed for the component/s	

4.1.4 Test case : CarMode Dyno (Ver: 6)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10681
Legacy ID:
Test Environment:

Test case Description:

Purpose:
Objective:

Test the functionality and quiescent current consumption in CarMode = DYNO

Test strategy:

preconditions and information:

- **All measurements shall be started when the component is powerfeed.**

  When the current is stable the actual measurement shall take place; the average and maximum value shall be evaluated during one minute.

  The time for stable current consumption varies for different components and modes and may in some cases take up to 30 minutes.

- All loads connected to the ECU, e.g. external sensors, shall be included during measurements.

Requirements:

Test Script Description :

Step	Action	Expected Result
1	Powerfeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarMo dSubtyp = Normal {0b000}	CarMode Normal;Normal is sent out on Network
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on Network The tested ECU shall be set in required internal

Step	Action	Expected Result
		mode due to CarMode Normal;Normal UsageMode Active
6	Set CarMode Dyno;Dyno2 CarModSts1 = Dyno {0b101}, CarModSubtypWdCarMo dSubtyp = Dyno2 {0b000}	CarMode Dyno;Dyno2 is sent out on the network The tested ECU shall be set in required internal mode due to CarMode Dyno;Dyno2 UsageMode Active
7	Set UsageMode to Driving UsgModSts = Driving {0b1101}	UsageMode Driving is sent out on the Network. The tested ECU shall be set in required Internal mode due to CarMode Dyno;Dyno2 Usage Mode Driving
8	Fill in if the ECU is in Sleep, Standby or Awake state in the attached excel template under the Sleep/awake sheet.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
9	Measure the current consumption. (Measure in CarMode Dyno;Dyno2 & UsageMode Driving) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
10	Set CarMode to Dyno;Dyno4 CarModSts1 = Dyno {0b101}, CarModSubtypWdCarMo dSubtyp = Dyno4 {0b001}	CarMode Dyno;Dyno4 is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Dyno;Dyno4 UsageMode Driving
11	Measure the current consumption. (Measure in CarMode Dyno;Dyno4 & UsageMode Driving) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
12	Check DTC to see if any is set due to this test	No DTC's are set
13	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
14	Check for errorframes and fill in the amount of error frames detected in the attached excel template under the errorframe sheet. Fill in 0 if no were	The number of errorframes observed shall be in the attached excel template

Step	Action	Expected Result
	detected.	
15	Remove powerfeed for the Component/s	

4.1.5 Test case : CarMode Crash (Ver: 5)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10682
Legacy ID:
Test Environment:

Test case Description:

Purpose:
Objective:

Test the functionality and quiescent current consumption in CarMode = CRASH

Test strategy:

Preconditions and information:

- All loads connected to the ECU, e.g. external sensors, shall be included during measurements.

Requirements:

Test Script Description :

Step	Action	Expected Result
1	Powerfeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarMo dSubtyp = Normal {0b000}	CarMode Normal;Normal is sent out on Network
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on Network The tested ECU shall be set in required internal mode due to CarMode Normal;Normal UsageMode Active
6	Set Usage Mode to Driving UsgModSts = Driving	UsageMode Driving is sent out on the Network The tested ECU shall be

Step	Action	Expected Result
	{0b1101}	set in required internal mode due to CarMode Normal;Normal UsageMode Driving
7	Set CarMode to Crash;Crash Level 1 CarModSts1 = Crash {0b011} CarModSubtypWdCarModSubtyp = Crash Level 1 {0b000}	CarMode Crash;Crash level 1 is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Crash;Crash Level 1 UsageMode Driving
8	List the state of the functions that are affected in the attached excel template under Crash sheet.	The affected functions realized by this ECU that are affected by CarMode Crash is noted in the attached excel template
9	Set UsageMode to Active UsgModSts=Active {0b1011}	UsageMode Active is sent out on the Network The tested ECU shall be set in required internal mode due to CarMode Crash;Crash level 1 UsageMode Active
10	Check DTC to see if any is set due to this test	No DTC's are set
11	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
12	Check for errorframes and fill in the amount of error frames detected in the attached excel template under the errorframe sheet. Fill in 0 if no were detected.	The number of errorframes observed shall be in the attached excel template
13	Remove Powerfeed for the component/s	

4.1.6 Test case : Energy Consumption During vehicle Start (Ver: 5)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 10684
Legacy ID:
Test Environment:

Test case Description:

Preconditions and information:

- **All measurements shall be started when the component is powerfeed.**

  When the current is stable the actual measurement shall take place; the average and maximum value shall be evaluated during one minute.

  The time for stable current consumption varies for different components and modes and may in some cases take up to 30 minutes.
- All loads connected to the ECU, e.g. external sensors, shall be included during measurements.

Requirements:

Test Script Description :

Step	Action	Expected Result
1	Powerfeed the component	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarMod Subtyp = Normal {0b000}	CarMode Normal;Normal is sent out on Network
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on Network The tested ECU shall be set in required internal mode due to CarMode Normal;Normal UsageMode Active
6	Set ElPowerLevel to "12V cold start" PwrLvlElecMai = 0x1 PwrLvlElecSubtyp = 0x0	PwrLvlElecMai = 0x1 and PwrLvlElecSubtyp = 0x0 is sent out on the network

Step	Action	Expected Result
7	Measure the current consumption. (Measure in CarMode Normal;Normal & UsageMode Active, Component in standby) Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption shall be included in in the attached excel template.
8	Fill in Available or Not Available in the attached excel template under the ElpowerLevel Sheet. (Availability is refers to if the ECU limit functionality or not based on the PwrLvlElecMai och PwrLvlElecSubtyp signals.)	Available/Not Available is filled in the attached excel template.
9	Check DTC to see if any is set due to this test	No DTC's are set
10	Deactivate the PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
11	Remove Powerfeed for the components/s	

References in SystemWeaver model

Test scope: ECU recovery mechanism

Version:(3) Status:Released

Test specification summary	Ver	Status
ECU recovery mechanism	(3)	Released

4.2 Test Specification: ECU recovery mechanism (Ver: 3)

Test Specification Description :

4.2.1 Test case : Test for not wake up sources Pin XX (Ver: 3)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 442658
Legacy ID:
Test Environment:
Test Method:

Test case Description:
Test Strategy:

It is a general test steps, and all the Pins related to the ECU need to test as the general test steps.
This test case should be test both in car mode=Normal ,Transport, Factory,each HW Pin must be test once, and the test result need be recorded in the attached excel template.

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Switch off network emulation. Switch on power supply. If the network needs a condition to go to sleep, fulfil it. Wait the DUT go to sleep	Confirm the network is in sleep
2	Input an valid HW wake up signal to ECU HW Pin XX that is not wake up source(based on ECU of schematic input) 对ECU接插件上的每一个非唤醒源PIN脚，输入有效的硬线唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
3	Measure the quiescent current	The quiescent current consumption should be the same or

Step	Action	Expected Result
	consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
4	Take away the HW signals input	

4.2.2 Test case : HW input wake up sources Pin XX (Ver: 3)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 442659
Legacy ID:
Test Environment:
Test Method:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps. This test case should be test both in car mode=Normal , Transport, Factory,each invalid and valid HW wake up source must be test once, and the test result need be recorded in the attached excel template.

Explain:
signal A, signal B, signal C and signal N are the ECU HW input signals, the signal A is the ECU wake up signal, the others HW signal not are wake up sources. At as below test steps “Input an invalid wake up signal value “means the signal A have a valid signal type value (e.g. high level/low level/ fixed duty cycle PWM/ differential signaling and others type) to wake up ECU, but others invalid signal type values shouldn’t wake up ECU, when we testing this step should input each invalid signal type value to test. The invalid values should be evaluated according to the following rule:

Rule	Input Type	Valid equivalence class	Invalid equivalence class
Rule 1	Defined the value range (a, z) or defined the number of values x	Have 1 value:1 value in parentheses	Have 2 values:2 values outside parentheses
Rule 2	Single condition A	Have 1 value: A	Have 1 value: Not A
Rule 3	Boolean Type	Have 1 value: True	Have 1 value: False
Rule 4	Gather Type{a1,a2,…an}	Have N values: a1 or a2,…or an	Have 1 value: Not belong{a1,a2,…an}
Rule 5	Satisfy multiple conditions A1,A2…An	Have 1 value: A1&A2…&An	Have N values: $\overline{A1}$, $\overline{A2}$, … $\overline{An}$

For example: Signal A is a level signal wake up source, when input a high level signal to IO1,the ECU will wake up, accord with the rule table of rule 1, the invalid values as follow:

Input signal	Input Type	ECU test conditions	Valid equivalence class	Invalid equivalence class
Signal A	Level signal	ECU sleep status	High level	Low level ,Dangling
		ECU activate status	High level	Low level ,Dangling

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Switch off network emulation. Switch on power supply. If the network needs a condition to go to sleep, fulfil it. Wait the DUT go to sleep	Confirm the network is in sleep
2	Input an invalid HW wake up signal value to ECU HW Pin XX that is wake up source (accord with the rule table get the invalid value) 对ECU接插件上的唤醒源PIN脚，输入无效的硬线唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
3	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
4	Take away the HW signals input of test step 2, keeping the ECU power feed, wake up the ECU network(by simulation PNC activate or other ways), let the ECU start network communication	ECU wake up and network communication
5	Input the invalid HW wake up signal value to ECU HW Pin XX that is same with step2. Then stop simulation let the ECU network stop communication 对ECU接插件上的唤醒源PIN脚，输入无效的硬线唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
6	Measure the current consumption. Fill in the measured value in the attached excel template under the current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
7	Take away the HW signals input of test step5, keeping the ECU power feed, wake up the ECU network(by simulation PNC activate or other ways), let the ECU start network communication.	ECU wake up and network communication
8	Input an valid HW wake up signal value to ECU HW Pin XX that is same with step2. Then stop simulation PNC activate, network communication stop, keeping the valid signal input more than 30 mins. (30mins according to the ECU fault mechanism definition) 对ECU接插件上的唤醒源PIN脚，持续30mins以上输入有效的硬线唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
9	Measure the current consumption. Fill in the measured value in the attached excel template under the current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
10	Input the valid HW wake up signal value to ECU HW Pin XX that is same with step2. then move it.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.

Step	Action	Expected Result
	对ECU接插件上的唤醒源PIN脚，输入有效的硬线唤醒信号。	
11	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

4.2.3 Test case : Internal timer wake up sources Timer XX (Ver: 3)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 442660
Legacy ID:
Test Environment:
Test Method:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test both in car mode=Normal , Transport, Factory,each Internal timer wake up source must be test once, and the test result need be recorded in the attached excel template.

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Switch off network emulation. Switch on power supply. If the network needs a condition to go to sleep, fulfil it. Wait the DUT go to sleep	Confirm the network is in sleep
2	keeping the ECU power feed, wake up the ECU network(by simulation PNC activate or other ways), let the ECU start network communication	ECU wake up and network communication
3	Set an internal timer XX (fulfil all activate conditions)	
4	After internal timer start, interrupt timer by external condition	
5	Stop simulation, let the DUT network stop communication	ECU network stop communication
6	Measure the current consumption. Fill in the measured value in the attached excel template under the current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
7	keeping the ECU power feed, wake up the DUT network(by simulation PNC activate or other ways), let the DUT start network communication	ECU wake up and network communication
8	Set an internal timer XX (fulfil all activate conditions)	
9	Stop simulation, let the DUT network stop communication	ECU network stop communication

Step	Action	Expected Result
10	Let the timer XX timeout	
11	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

4.2.4 Test case : Sensor signal input wake up sources Signal XX (Ver: 3)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 442661
Legacy ID:
Test Environment:
Test Method:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test both in car mode=Normal,Transport, Factory,each invalid and valid sensor signal wake up source must be test once, and the test result need be recorded in the attached excel template.

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Switch off network emulation. Switch on power supply. If the network needs a condition to go to sleep, fulfil it. Wait the DUT go to sleep	Confirm the network is in sleep
2	Input an invalid Sensor wake up signal value to ECU sensor Pin XX that is wake up source (accord with the rule table get the invalid value) 对ECU接插件上的传感器唤醒源PIN脚，输入无效的传感器唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
3	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
4	Take away the HW snesor signals input of test step 2, keeping the ECU power feed, wake up the ECU network(by simulation PNC activate or other ways), let the ECU start network communication	ECU wake up and network communication
5	Input an invalid Sensor wake up signal value to ECU sensor Pin XX that is same with step2. Then stop simulation let the DUT network stop communication 对ECU接插件上的传感器唤醒源PIN脚，输	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.

Step	Action	Expected Result
	入无效的传感器唤醒信号。	
6	Measure the current consumption. Fill in the measured value in the attached excel template under the current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
7	Take away the HW sensor signals input of test step5, keeping the ECU power feed, wake up the ECU network(by simulation PNC activate or other ways), let the ECU start network communication, Set UsageMode to Convenience UsgModSts=Convenience {0b0010}	ECU wake up and network communication, UsageMode Convenience is sent out on Network.The DUT shall be set in required internal mode due to CarMode Normal
8	Input an valid Sensor wake up signal value to ECU sensor Pin XX that is same with step2. Then stop simulation PNC activate, network communication stop, keeping the valid signal input more than 30 mins. 对ECU接插件上的传感器唤醒源PIN脚，持续30mins以上输入有效的传感器唤醒信号。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
9	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
10	Input an valid Sensor wake up signal value to ECU sensor Pin XX that is same with step2. . then move it. 对ECU接插件上的唤醒源PIN脚，输入有效的硬线唤醒信号,之后移除。	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
11	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

4.2.5 Test case : Wireless wakeup sources Wireless XX (Ver: 3)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 442662
Legacy ID:
Test Environment:
Test Method:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test both in car mode=Normal , Transport, Factory,each sensor signal wake up source must be test once, and the test result need be recorded in the attached excel template.

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Switch off network emulation. Switch on power supply. If the network needs a condition to go to sleep, fulfil it. Wait the DUT go to sleep	Confirm the network is in sleep
2	Inject same frequency wireless wakeup tiger XX	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
3	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
4	Send SMS message disturb	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
5	Measure the quiescent current consumption. Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
6	Inject valid wireless wakeup tiger XX, keeping the valid signal input more than 30 mins.	Value Awake, Standby or Sleep dependent on ECU behaviour is filled in the attached excel template.
7	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement

Step	Action	Expected Result
	Fill in the measured value in the attached excel template under the quiescent current consumption sheet.	from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

4.2.6 Test case : ECU recovery mechanism when High and Low Power supply (Ver: 2)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 850807
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Set the power supply to 12 V (+/-0.1 V) and Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
2	Connect the external trigger signal(s) such that the DUT communicates continuously.	The DUT communicates with the Network Analyser.
3	Set the power supply to 8 V (-0.0/+0.1 V). Repeat steps 2 and 3.	Repeat steps 2 and 3.
4	Monitor the bus using the oscilloscope and Network Analyser. Set the power supply to 8 V. Trigger the module to communicate continuously. Slowly reduce the power supply voltage (no faster than 1 V/sec) until the DUT ceases communication. Record the voltage at which the DUT ceased communication.	The module shall 'fail silent', i.e. not transmit invalid frames, when the power supply drops below the minimum transmission voltage. Note: this includes dominant pulses during module reset. Record the voltage at which communication ceased as Vstop_low. Vstop_low < 8 V.
5	Following step 4, with the trigger signal(s) still active, slowly (no faster than 1 V/s) increase the power supply voltage until the module begins to communicate continuously. Use the network analyser and oscilloscope to monitor for invalid frames and/or dominant spikes on the network as the module starts communication. Record the voltage at which communication commenced.	The module shall recommence correct communication at less than or equal to 8 V and 'fail silent', as defined in step 5, below the minimum transmission voltage. Record the voltage at which communication began as Vcomm_low. Vcomm_low < 8 V.
6	Set the power supply to 12 V (+/-0.1 V) ,disconnect the external trigger signal(s) , Measure the current consumption.	The current consumption should be the same or lower than Current consumption requirement from VMM on this ECU. The measured current consumption result shall be included in the attached excel template.
7	Set the power supply to 16 V (-0.1/+0.0 V). Repeat steps 2 and 3.	As steps 2 and 3.

Step	Action	Expected Result
8	Monitor the bus using the oscilloscope and Network Analyser. Set the power supply to 16 V. Trigger the module to communicate continuously. Slowly increase the power supply voltage (no faster than 1 V/sec) until the module ceases communication or up to 24 V. Record the voltage at which the module ceased communication.	The module shall 'fail silent', i.e. not transmit invalid frames, when the power supply drops above the maximum transmission voltage. Note: this includes dominant pulses during module reset. Record the voltage at which communication ceased as Vstop_high. Vstop_high > 16 V. It is acceptable that application messages stop before NM message, and vice-versa.
9	Set the power supply to 12 V (+/-0.1 V) ,disconnect the external trigger signal(s) , Measure the quiescent current consumption	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

4.2.7 Test case : ECU shoud have timeout mechanism and ECU should reset when Vote and PNC can not set 0 on time (Ver: 2)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 850808
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.

Note:this test case used for all ECU with NM frame which have vote and PNC

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
2	Clear all DTC	
3	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000}	CarMode Normal; Normal is sent out on the Networks.
4	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
5	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
6	Set Vote=1, PNC=0 in ECU SW and simulation of test environment signals on the bus	the ECU timer should increase
7	wait 15s	the ECU timer should be 0 and ECU should reset
8	Deactivate the Vote,PNC inECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
9	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU.

Step	Action	Expected Result
		The measured quiescent current consumption result shall be included in the attached excel template.
10	Set Vote=0, PNC=1 Iin ECU SW and simulation of test environment signals on the bus	the ECU timer should increase
11	wait 15s	the ECU timer should be 0 and ECU should reset
12	Deactivate theVote, PNC in ECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
13	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
14	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 1-13	As repeat step 1-13
15	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 1-13	As repeat step 1-13
16	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
17	Clear all DTC	
18	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000}	CarMode Normal; Normal is sent out on the Networks.
19	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
20	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
21	Set Vote=0, PNC=0 in ECU SW and simulation of test environment signals on the bus	the ECU timer should be 0
22	Deactivate the Vote,PNC in ECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
23	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
24	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 16-23	As repeat step 16-23
25	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 16-23	As repeat step 16-23
26	Set CarMode Transport; Transport CarModSts1 = Transport {0b001}	As repeat step1-25

Step	Action	Expected Result
	CarModSubtypWdC arModSubtyp = Transport {0b000} repeat step1-25	
27	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubt yp = Factory {0b000} repeat step1-25	As repeat step1-25
28	Remove Powerfeed for the Component/s.	
29		

4.2.8 Test case : ECU shoud have timeout mechanism and ECU should reset when RG (Resource group), PNC can not set 0 (Ver: 2)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 850809
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.
Note:this test case used for all RG and PNC of the ECU

Requirements:

Requirement	Req ID	Ver	ASIL	Status
Network/Component failsafe timeout	98076	(3)		CS Released

Test Script Description :

Step	Action	Expected Result
1	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
2	Clear all DTC	
3	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000}	CarMode Normal; Normal is sent out on the Networks.
4	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
5	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
6	Set RG=!0 PNC=0 and simulation of test environment signals on the bus	the ECU timer should increase
7	wait 3min	the ECU timer should be 0 and ECU should reset
8	Deactivate the RG ,PNC and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
9	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included

Step	Action	Expected Result
		in the attached excel template.
10	Set RG=0 PNC=1 in ECU SW and simulation of test environment signals on the bus	the ECU timer should increase
11	wait 3min	the ECU timer should be 0 and ECU should reset
12	Deactivate the RG,PNC in ECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
13	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template
14	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 1-13	As repeat step 1-13
15	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 1-13	As repeat step 1-13
16	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
17	Clear all DTC	
18	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubt yp = Normal {0b000}	CarMode Normal; Normal is sent out on the Networks.
19	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
20	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
21	Set RG=0 PNC=0 and simulation of test environment signals on the bus	the ECU timer should be 0
22	Deactivate the RG,PNC in ECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
23	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.
24	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 16-23	As repeat step 16-23
25	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 16-23	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 16-23
26	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdC	As repeat step 1-25

Step	Action	Expected Result
	arModSubtyp = Transport {0b000} repeat step 1-25	
27	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubt yp = Factory {0b000} repeat step1-25	As repeat step1-25
28	Remove Powerfeed for the Component/s.	

4.2.9 Test case : ECU shoud have timeout mechanism when NM frame is sent with PNC is 0 more than 15s (Ver: 1)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 1324216
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.

Note:this test case used for all PNC of the ECU

Requirements:

Test Script Description :

Step	Action	Expected Result
1	PowerFeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000	CarMode Normal; Normal is sent out on the Networks.
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
6	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
7	the ECU send NM frame with PNC=0 in ECU SW and simulation of test environment signals on the bus	the ECU timer should increase
8	wait 15s	the ECU timer should be 0 and ECU should reset
9	Deactivate the NM frame,PNC inECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
10	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in the attached excel template.

Step	Action	Expected Result
11	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 1-9	As repeat step 1-9
12	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 1-9	As repeat step 1-9
13	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
14	Clear all DTC	
15	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdC arModSubtyp = Transport {0b000} repeat step1-9	As repeat step1-9
16	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubt yp = Factory {0b000} repeat step1-9	As repeat step1-9
17	Remove Powerfeed for the Component/s.	

4.2.10 Test case : ECU shoud have timeout mechanism when ECU just send APP frame and do not send NM frame (Ver: 1)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 1324217
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.

Note:this test case used for all ECU with NM frame and APP frame

Requirements:

Test Script Description :

Step	Action	Expected Result
1	PowerFeed the component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000}, CarModSubtypWdCarModSubtyp = Normal {0b000	CarMode Normal; Normal is sent out on the Networks.
5	Set UsageMode to Active UsgModSts = Active {0b1011}	UsageMode Active is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal, Usage Mode Active.
6	Set UsageMode to Abandoned. UsgModSts = Abandoned {0b0000}	UsageMode Abandoned is sent out on the Network. The tested ECU shall be set in required internal mode due to CarMode Normal, UsageMode Abandoned.
7	the ECU just send APP frame and do not send NM frame in ECU SW and simulation of test environment signals on the bus	the ECU timer should increase
8	wait 15s	the ECU timer should be 0 and ECU should reset
9	Deactivate the APP frame in ECU SW and simulation of test environment signals on the bus.	The simulated network communication shall be deactivated.
10	Measure the quiescent current consumption.	The quiescent current consumption should be the same or lower than quiescent current consumption requirement from VMM on this ECU. The measured quiescent current consumption result shall be included in

Step	Action	Expected Result
		the attached excel template.
11	Set UsageMode to inactive. UsgModSts = inactive {0b0001} repeat step 1-9	As repeat step 1-9
12	Set UsageMode to convenience. UsgModSts = convenience {0b0010} repeat step 1-9	As repeat step 1-9
13	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
14	Clear all DTC	
15	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdC arModSubtyp = Transport {0b000} repeat step1-9	As repeat step 1-9
16	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubt yp = Factory {0b000} repeat step1-9	As repeat step 1-9
17	Remove Powerfeed for the Component/s.	

References in SystemWeaver model

Test scope: VMM ECU Reset

Version:(1) Status:Released

Test specification summary	Ver	Status
ECU Reset	(1)	Released

4.3 Test Specification: ECU Reset (Ver: 1)

Test Specification Description :

4.3.1 Test case : Master ECU reset when Quiescent current monitor is active (Ver: 1)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 1114497
Legacy ID:
Test Environment:

Test case Description:
This DVM is only applicable for ECUs who are activators of Reset
Purpose:
Testing the ECU reset function when the Quiescent current monitor is active with car mode is Normal or Factory or Transport, usagemode is abandoned, the ECU can reset itself and other ECU according to send command.
Test Strategy:
The component shall be powerfeed with a stable power supply.
All loads connected to the ECU, e.g. external sensors, shall be connected to the ecu during the test.
Make sure to simulate the CarTiGlb signal and to include it in the steps where the network communication is active.
The CarTiGlb is required to analyze the QCM DID.

PreCondition:

Car Mode: Normal
PNC29=0
Usage Mode: Inactive

DID D905 is set to ECU Rest Activated
Notes:
1.the DVM used for GEEA2.0 VGM/BGM;
2.the DVM used for GEEA3.0 DHU

Requirements:

Test Script Description :

Step	Action	Expected Result
1	PowerFeed the Component/s	
2	Activate the network communication by setting a PNC which the signal VehModMngtGlbSafe1 is mapped to and that the ECU is a member of to simulate the test environment signals on the bus.	The network communication is active.
3	Clear all DTC	
4	Set CarMode Normal CarModSts1 = Normal {0b000} CarModSubtypWdCarModSubtyp = Normal {0b000}	CarMode Nornal; Normal is sent out on the network.
5	Change UsageMode to a UsageMode where your ECU is allowed to activate a PNC.	The UsageMode status is is sent out on network. The tested ECU shall be set in required internal mode due to CarMode Normal and the chosen UsageMode
6	Deactivate the VFC/PNC and simulation of test environment signals on the bus.	The tested ECU shall be set in required internal mode due to CarMode Normal and the chosen UsageMode The network communication shall be deactivated.
7	Wait 30 sec	
8	Activate the network communication from the ECU in test by setting a VFC/PNC which the ECU is an activator of. If the VFC activated have a timeout delay. Reactivate the VFC continously to keep the network communication active.	The network communication is active and the ECU in test should be the component keeping the network active.
9	Wait at least 15 seconds since the previous step and then set the VFC/PNC used for distributing the VMM signal group globally and simulate the test environment signals on the bus.	The network communication is expanded with the VMM signalling.
10	Set signal VehModMngtGlbSafe1.FltEgyCns = FltEgyCns1_Flt Wait 14 seconds Set signal VehModMngtGlbSafe1.FltEgyCns = FltEgyCns1_NoFlt	The signal VehModMngtGlbSafe1.FltEgyCns is distributed on the bus.
11	1. the ECU send and receive diag 0x11 0x81 2. Change UsageMode to to Inctive	the ECU is reset
12	3. Change UsageMode to to Abandoned	2. UsageMode Abandoned is now sent out on the network.
13	Deactivate all communication. Both VFC/PNC activated in the ECU and in the simulated environment,	
14	Set CarMode Transport; Transport CarModSts1 = Transport {0b001} CarModSubtypWdC arModSubtyp = Transport {0b000}	CarMode Transport is sent out on the Network
15	repeat step5-13	the ECU is reset
16	Set CarMode Factory;Factory CarModSts1 = Factory {0b010}, CarModSubtypWdCarModSubtyp = Factory {0b000}	CarMode Factory is sent out on the network
17	repeat step5-13	the ECU is reset
18		

References in SystemWeaver model

Test scope: VMM ECU DID

Version:(1) Status:Released

Test specification summary	Ver	Status
VMM ECU DID	(1)	Released

4.4 Test Specification: VMM ECU DID (Ver: 1)

Test Specification Description :

4.4.1 Test case : VMM ECU DID (Ver: 1)

Version History:

Conditional values for the Test Case:

Testcase prio:

Test Case ID: 1324218
Legacy ID:
Test Environment:

Test case Description:
Test Strategy:
It is a general test steps，and all the Activators related to the ECU need to test as the general test steps.
This test case should be test in car mode=Normal ,Transport,Factory , and the test result need be recorded in the attached excel template.

Note:
1. this test case used for CEM/BGM for GEEA2.0;
2.this test case used for ZCU for GEEA3.0;

Requirements:

Test Script Description :

Step	Action	Expected Result
1	PowerFeed the component/s	
2	read the flowing genernal information DID with service 22 1. F1AE (ECU Software Part Numbers - Geely) 2.F1AA (ECU Core Assembly Part Number - Geely) 3.F1AB (ECU Delivery Assembly Part Number - Geely) 4.F18C (ECU Serial Number) 5.F106 (Vehicle Configuration Parameters)	the ECU should give positive response
3	read the flowing VMM information DID with service 22 1.E503 (VFCvectorFrame switch) 2.DD00(Global Real Time) 3.D005 (High Quiescent Current Fault Identification	the ECU should give positive response

Step	Action	Expected Result
	Data) 4.D999 (High Quiescent Current Fault Identification Data) 5.F010 (CEM Wake Up Cause) 6.E500 (Reason for long time in Usage Mode INACTIVE) 7.DD0A (UM) 8.D134 (CM) 9.430E (Usage Mode Extention Statistics) 10.430F (Usage Mode Time Statistics) 11.4261 (Remote key history) 12.430E (U-M stay time) 13.42F6 (Braking status) 14.42FC (Brake pedal duty cycle) 15.430F (IHU different UM stay times)	
4	read the flowing 12 battery information DID with service 22 1.4028 (Vehicle Battery State Of Charge - Estimated) 2.42D1 (Battery Internal Resistance Statistics Counter) 3.42CD (Battery Average Quiescent Current - Statistics) 4.40D7 (Battery Quiescent Current - Statistics) 5.4090 (12V battery current value) 6.4029 (12V battery temperature) 7.41A9(Stop - Start Feature Inhibit Reasons) 8.42D2 (Battery Normalized Inter Resistance) 9.42D3(Battery Internal Resistance)	the ECU should give positive response
5	send 19 02 20 to read DTC	the ECU should give positive response
6	send 11 01 to rest ECU	the ECU should be rest
7	Remove Powerfeed for the Component/s.	
8		

5 Analysis

TBD

6 Evaluation

TBD